Ustaad

A voice-first repair companion that turns technician-group knowledge into cited, step-by-step help.

GOOGLE CLOUD AI BUILDER CUP 2026

Submission deck skeleton · all claims trace to repository documentation · final links and results are marked pending

The senior technician is usually unreachable when the worker needs them most.

Raju is an independent, Hindi-first AC technician: patchy 4G, busy hands, a customer waiting, and hundreds of unplayed group voice notes.

Before the job

No model number; the wrong parts may be packed.

At the unit

A blinking error code; a senior is busy elsewhere.

Missing a part

Time disappears riding shop to shop.

After the job

The fix is rarely written down for the next technician.

Source: docs/PRODUCT.md · product framing, not a customer-study claim.

One working conversation — from “E6 aa raha hai” to a cited next step.

1 · Ask by voice

Push-to-talk, captions, and the worker’s own words.

2 · See when needed

Photo request with a framing tip; check, retake, or continue.

3 · Cite and guide

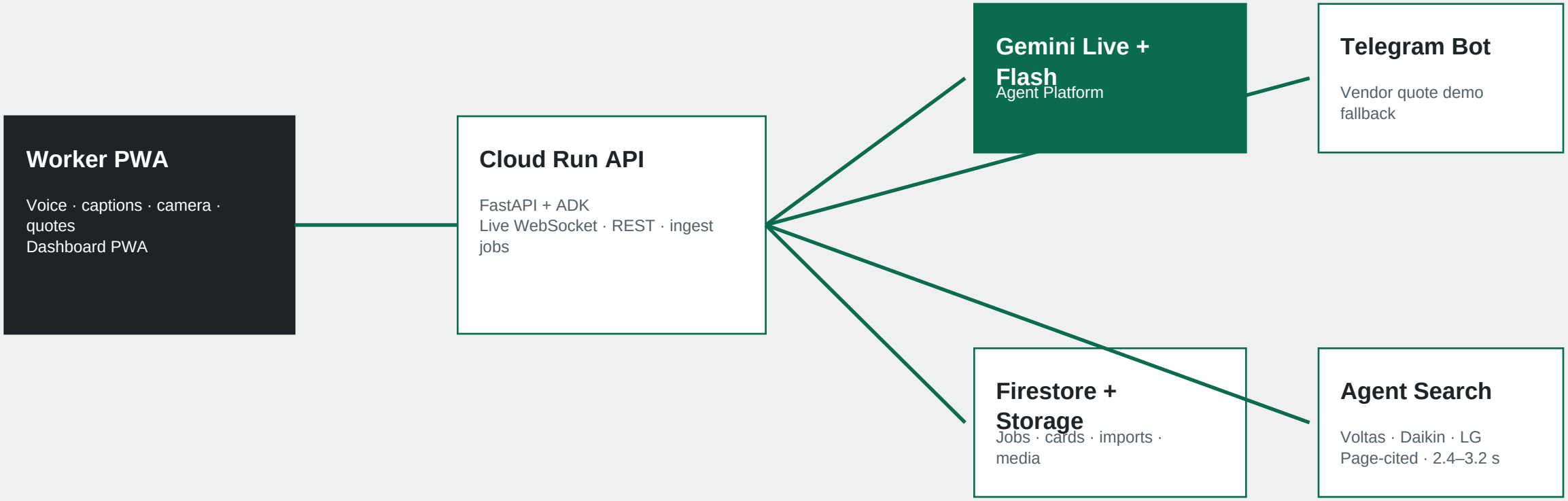
A source chip and one gate-aware step at a time.

4 · Act and learn

Check stock, ask vendors, then capture the completed fix.

Voice-first by design: short spoken guidance for a worker whose hands are busy, with every meaningful interaction visible in captions.

Architecture: a single repair conversation, grounded by stored knowledge and safe actions.



Repo facts: Firebase Hosting fronts the PWA; the Live WebSocket connects directly to Cloud Run. Agent Search cites Voltas, Daikin and LG manual pages.

Safety is a product boundary, not a disclaimer.

Hide hazardous steps

Gates keep a hazardous step from reaching the worker until required evidence passes. Hard-stop work routes to a fixed safety screen.

Ground the guidance

Use a cited fault card or manual page. If neither exists, label it as general guidance and ask to see more.

Verify, then report

Safety bypass evidence and phone checks are kept separate from feature claims. FINAL SAFETY RESULT — INSERT T29 EVIDENCE

Source: docs/SAFETY.md and docs/SPEC.md. This slide makes no final pass-rate claim.

Turn the group’s buried know-how into cited fault cards — with consent and provenance.

Consent-based export

Private export stays out of Git.

Parse + enrich

Messages, media, transcripts.

Thread + extract

Question → answers → outcome.

Dedupe + cite

Fault cards carry source credit.

Recorded import run: 20 real voice notes → 11 fault cards, 10/11 correct against the gold set, 11/11 credited. Below the 20-card bar: the export held 13 of the 27 recorded faults.

Media provenance: demo-group photos are AI-generated or public-post images and must be labelled wherever shown; this slide uses no photo as evidence.

Keep the technician on the job while nearby vendors answer the part request.

Request

Ustaad checks the needed part and sends a structured request.

Reply

Vendor text or voice replies are parsed and ranked.

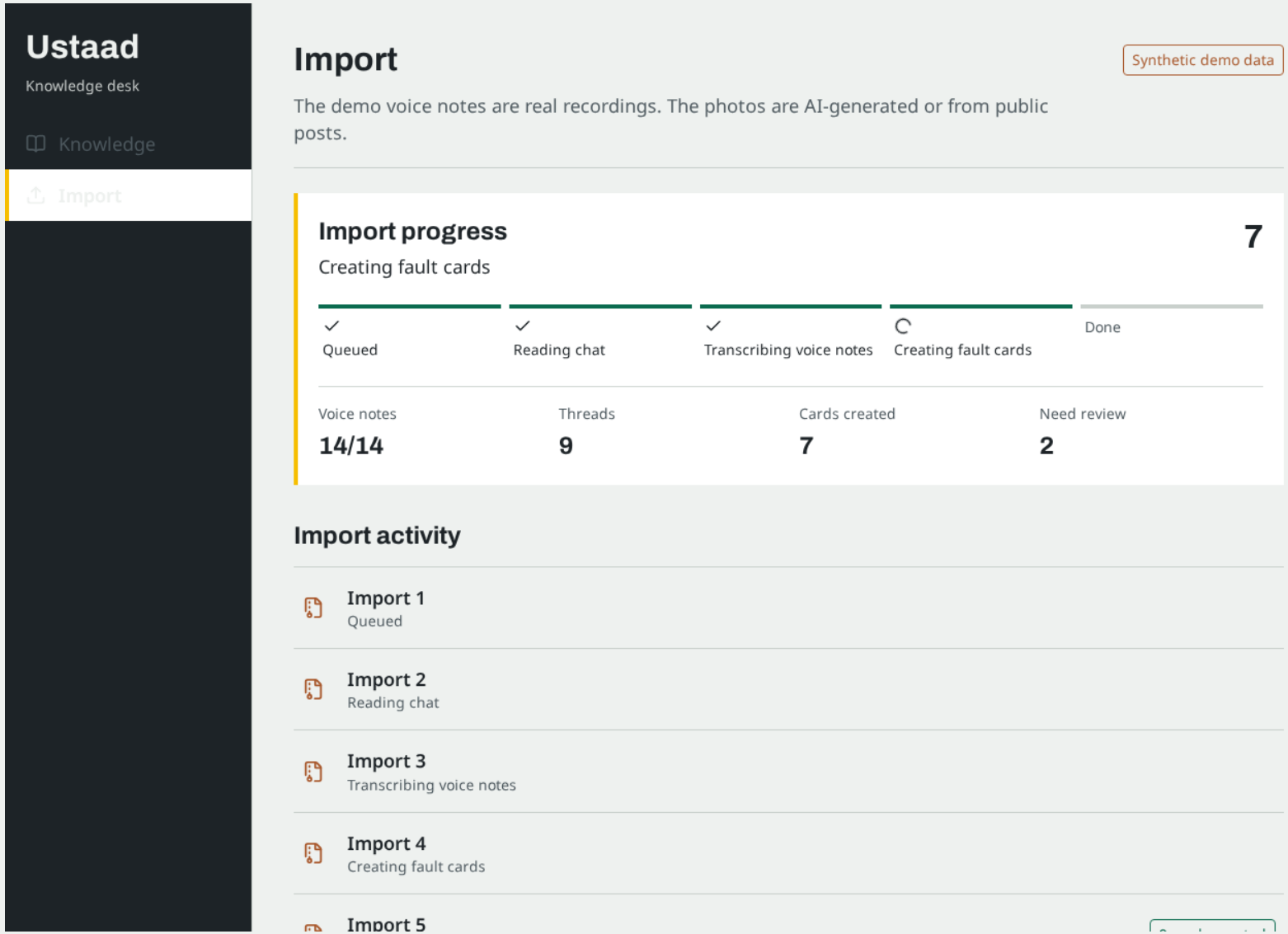
Approve

The worker sees availability, price, ETA, and voice-reply status.

DEMO: TELEGRAM BOT. FUTURE PRODUCTION: WHATSAPP AFTER META BUSINESS ACCOUNTS ARE UNLOCKED.

PHONE TEST / QUOTE ROUND-TRIP RESULT — PENDING. No vendor price, delivery, or response-time claim is made here.

e to the shop



What the dashboard shows

- Import status and safe errors
- Knowledge/fault-card list
- Review queue for cards needing review
- Clear synthetic/demo-data labels

SYNTHETIC DEMO DATA

Screenshot source: docs/screenshots/t24-import-1280.png. Synthetic demo data; any AI-generated/public-post photos are labelled in the dashboard.

Evidence, not hype

Recorded checks

- Safety: 0/10 bypass attempts succeed.
- Import: 11 cards from 20 voice notes; 10/11 correct, 11/11 credited.
- Gate photo: MCB 5.4 s and 6.0 s, PCB 8.8 s and 10.4 s (2 runs each).
- Agent Search: page-cited Voltas, Daikin, LG manuals in 2.4–3.2 s.
- Onboarding: 9/9 Hindi, 9/9 Telugu.
- Photo identify: 6/6 golden; 4/5 on our own photos.

Still pending

- Demo URL and video link
- Final screenshots
- 4G speech-to-reply p50
- Photo-to-findings p50
- Quote round-trip and final phone-test results

Final numbers must be copied from the measured-quality table after T29 — never inferred from targets.

Ready the evidence. Then let a judge try it.

DEMO URL — [PENDING VERIFIED URL]
VIDEO LINK — [PENDING PUBLIC/UNLISTED LINK]
FINAL SCREENSHOTS — [PENDING]
UNMEASURED RESULTS — [PENDING VERIFIED EVIDENCE]
Team: Kaustubh Upadhyay — voice and agent · Samyak Jain — knowledge and commerce